RAJ B PRAJAPATI

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Backend & Distributed Systems Engineer with 5+ years shipping production platforms across power-grid simulation, IoT, and bioinformatics. Multi-cloud certified (AWS Solutions Architect, GCP Professional Cloud Developer, GCP Professional Data Engineer). Led cross-functional teams of up to 8 engineers end-to-end — from architecture through post-launch reliability — cutting infrastructure cost by 80% and scaling simulation throughput 10x.

CERTIFICATIONS

- AWS Solutions Architect Certification — Amazon Web Services, July 2026
- GCP Professional Data Engineer Certification — Google Cloud Platform, December 2025
- GCP Professional Cloud Developer Certification — Google Cloud Platform, January 2025
- Microsoft Technology Associate – Python 3 — Microsoft, 2020

WORK EXPERIENCE

Software Engineer 3 **Jul 2025 — Present**
Infocusp Innovations LLP *Ahmedabad, India*

- Currently leading a team of 8 full-stack developers on a new AI-based learning and development project, owning ideation, scoping, milestone definition, and task management end-to-end.
- Received the company-wide Appreciation for Q2 deliverables, recognized for standout performance in model ingestion and benchmarking, and for consistently supporting the team across multiple workstreams.
- Led end-to-end delivery of a full-stack application as tech lead for a 3-engineer team, setting coding and review standards that kept 100% of committed features on schedule across two quarters.
- Architected an event-driven, auto-scaling simulation engine for power systems, lifting daily throughput from 100+ to 1,000+ simulations/day (10x) without added infrastructure cost.
- Led a zero-downtime upgrade of a production Elasticsearch cluster to the latest major version with zero data loss, sustaining 99.99% availability throughout.
- Redesigned Elasticsearch indexing strategy and Firebase I/O patterns, cutting Firebase operational cost by ~80% while improving query latency.

Software Engineer 2 **Nov 2022 — Jul 2025**
Infocusp Innovations LLP *Ahmedabad, India*

Power Systems & Grid Planning

- Built a large-scale grid-planning and decarbonization platform, powering advanced simulation and analysis workflows used across engineering teams.
- Designed a high-throughput Kotlin data-ingestion pipeline parsing and normalizing CIM/GML power-grid formats into a unified internal model, eliminating manual data prep.
- Automated simulation processing end-to-end via a pre/post-processing framework integrated with GridLAB-D, scaling capacity from 0 to 100+ simulations/day.
- Deployed containerized serverless workloads on GCP for N-1 contingency analysis and benchmarking, reducing idle infrastructure spend.

Bioinformatics Infrastructure

- Built backend services and batch jobs processing 50+ GB of biological sequence data per day at >90% test coverage, cutting production defects significantly quarter over quarter.
- Owned operational workflows (monitoring, retries, alerting) for the batch pipeline, reducing failed-job recovery time and improving on-call reliability.

IoT Smart Mattress Platform

- Led a cross-functional team of 8 engineers across frontend, backend, and cloud to take the platform from prototype to production.
- Drove sprint planning, stakeholder alignment, and third-party integrations, accelerating feature adoption and shortening release cycles.
- Mentored interns and junior engineers, cutting new-hire onboarding time and lifting team velocity.
- Authored and executed system migration and architecture proposals that improved scalability with minimal production disruption.

Software Engineer 1 **Jun 2021 — Nov 2022**
Infocusp Innovations LLP *Ahmedabad, India*

- Built and shipped scalable REST APIs and web applications with React.js and Python on AWS, sustaining 99.9% uptime in production.

- Built CI/CD pipelines and Karate-based integration test suites, reaching 100% test coverage as a release gate before every production deploy.
- Designed backend subsystems for push notifications and scheduled jobs handling 500+ events/day across distributed services.
- Engineered and operated AWS cloud infrastructure for high availability and fault tolerance across production workloads.

Software Engineer Intern **Feb 2019 — Jun 2019**
MMSI Technologies Pvt. Ltd. *Ahmedabad, India*

- Built and maintained client projects in PHP, including static websites in HTML/CSS.
- Automated data migration using Python web scraping (BeautifulSoup), replacing manual data collection.

SKILLS

- Languages: Python, Kotlin, JavaScript, Java, HTML/CSS
- Cloud & DevOps: AWS, GCP, Docker, Terraform, AWS SAM, CI/CD
- Backend & APIs: REST, gRPC, GraphQL, Protobuf, FastAPI, Firebase
- Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Firestore
- Data & Messaging: RabbitMQ, MQTT, GCP Pub/Sub, AWS IoT Core, Apache Spark, Apache Airflow
- Frontend: React.js
- Testing & Tools: Pytest, Karate, Artillery, Git, Bazel
- Concepts: Distributed Systems, Microservices, Event-Driven Architecture, TDD

SELECTED PROJECTS

- NMT for Indian Languages (DA-IICT Research): Built transformer-based NMT systems for Gujarati-Hindi and Gujarati-English, achieving state-of-the-art BLEU scores for low-resource language pairs.
- Cellular Automata on CUDA: Implemented Game of Life (23/3 rule) on 2D/3D grids with GPU-parallel state validation, cutting computation time versus CPU baselines.

EDUCATION

M.Tech, Machine Learning — Dhirubhai Ambani Institute of Information and Communication Technology Aug 2019 — May 2021

- GPA: 8.0/10.0 · Teaching Assistant (Computer Networks, Discrete Mathematics) · NLP/Machine Translation research, Information Retrieval Lab

B.Tech — Parul University Aug 2015 — May 2019

- GPA: 9.02/10.0 · Academic Excellence Awards (Gold, Silver, Bronze) · ERASMUS+ fully-funded exchange, Wroclaw University of Science and Technology, Poland

LANGUAGES KNOWN

English · Hindi · Gujarati · German (A1.1)